Fine-Grained Mention-Level Analysis of Biomedical Entity Linking Models

Baptiste PRAS[1] and Nona NADERI
Université Paris-Saclay, CNRS, Laboratoire Interdisciplinaire des Sciences du Numérique, 91400, Orsay, France
ORCiD ID: Baptiste PRAS https://orcid.org/0009-0004-5780-1287
ORCiD ID: Nona NADERI https://orcid.org/0000-0002-1272-7640

Abstract. Biomedical Entity Linking (BEL) is essential for structuring knowledge from biomedical texts. While many BEL systems achieve high overall performance, global metrics like recall often mask significant variations in model behavior. In this work, we propose a fine-grained evaluation framework that analyzes performance across a set of mention-level characteristics, like length, lexical variation, synonymy, homonymy, and training frequency. Using the BELB benchmark, we apply our analysis to different neural BEL systems, including GenBioEL and ArboEL. Our results show that models behave differently: GenBioEL maintains robust performance on rare or ambiguous mentions, while ArboEL suffers severe performance degradation, especially on unseen and long mentions.

Keywords. entity linking, entity disambiguation, named entity, knowledge base.

1. Introduction

Biomedical Entity Linking (BEL) is the task of mapping mentions of biomedical concepts in text to standardized identifiers in a knowledge base, such as UMLS [1]. It plays a central role in biomedical natural language processing, enabling downstream applications like information retrieval, clinical decision support, and knowledge base construction.

Recent neural BEL systems have achieved strong global performance, but evaluation practices remain limited. Most evaluations rely on aggregate metrics, such as recall@1, which can obscure specific failure cases. In particular, performance often varies greatly depending on mention properties, such as lexical ambiguity, surface form variability, or training frequency. To address these limitations, prior work has proposed more fine-grained evaluations. Agrawal et al. [2] introduced mention-level annotations (e.g., frequency, string match) for clinical normalization tasks. Varma et al. [3] examined the impact of surface similarity, ambiguity, and frequency on accuracy. More recently, the BELB benchmark [4] unified evaluation across eight BEL datasets with consistent data processing and metrics. Building on these efforts, we propose a general analysis framework that can be integrated directly into BELB. We automatically annotate test mentions with interpretable features—mention length, lexical variation, synonymy, homonymy, and training frequency—and analyze model performance across these

[1] Corresponding Author: Baptiste PRAS, baptiste.pras@universite-paris-saclay.fr

dimensions. We apply our framework to two contrasting neural models: ArboEL [5], a dual-encoder architecture with coreference modeling via arborescence, and GenBioEL [6], a generative model that produces entity IDs token by token. Our experiments uncover model-specific weaknesses, provide insights into generalization, and offer practical tools to guide model improvement. To facilitate comparison with future work on this task, we release the source code and data of our experiments.[2]

2. Methods

2.1. Characteristics, Binning, and Evaluation Protocol

We define the characteristics of mentions and bin the mentions into categories:

Mention length: number of tokens in the mention, computed with BERT-base-uncased tokenizer. Mentions with ≤ 10 tokens are classified as short; otherwise, as long.

Lexical variation: whether the mention text appears exactly as one of the names of the gold concept in the corresponding dataset's knowledge base (KB). We compute the normalized Levenshtein distance between the mention and the closest synonym of the gold concept. Mentions are classified as low lexical variation (distance ≤ 0.1); otherwise, as high lexical variation.

Synonymy: number of distinct names (synonyms) for the gold concept in the KB. Con- cepts with ≤ 10 are considered low synonymy; otherwise, as high synonymy.

Homonymy: number of different concepts that share a close (normalized Levenshtein distance ≤ 0.1), or exact name in the dictionary. A mention is labeled as homonymous if its surface form is associated with at least more than one concept in the KB.

Zero-shot / Rare name: whether the exact mention string was rare or unseen during train-ing. E.g., "HIV" is considered rare if it appears only a few times (≤ 10 times), and zero-shot if never seen; otherwise, it is frequent.

Zero-shot / Rare entities: whether the gold concept itself was rare or unseen during train-ing. E.g., the concept "C0282687" for Ebola will be considered zero-shot if it never appears in training. Concepts seen ≤ 10 are classified as rare; otherwise, as frequent.

Each of these characteristics is computed automatically from the KB of the dataset and the training set. These characteristics reflect both the mention difficulty (i.e., form or ambiguity) and availability in the training data. We evaluate model performance across categories (e.g., short vs. long mentions, or seen vs. unseen names). This gives us a more detailed view of strengths and weaknesses compared to a global score.

2.2. Biomedical Entity Linking Models

Among BELB's neural approaches, we focus our analysis on two top-performing representative architectures: GenBioEL and ArboEL. *ArboEL* is a discriminative model based on a BERT encoder and is trained in such a way that correct mention-mention and mention-entity edges receive higher scores than the incorrect ones, enforcing global consistency. *GenBioEL*, on the other hand, is a generative approach that formulates entity linking as a sequence generation task. It uses a BART encoder-decoder architecture and is trained to generate the surface form of the correct entity token given a sentence

[2] https://github.com/baptistepras/BEL-Mention-Analysis

containing the mention. To efficiently constrain generation, GenBioEL restricts decoding to valid entity names by organizing candidate surface forms into a prefix-based structure, allowing only valid completions during generation. In addition, we include a rule-based model (RBES) to provide a non-neural reference. Each RBES is tailored to its entity type and relies on lexical rules and string-matching heuristics to link mentions to concept names in the corresponding KB.

2.3. Datasets and Evaluation Procedure

We evaluate the models on three biomedical entity linking datasets: Linnaeus [7], S800 [8], and MedMentions [9]. These datasets were selected for their diversity in domain coverage. All models are evaluated using BELB, which ensures consistency in evaluation by unifying the format of datasets, providing a common KB, and standardizing the evaluation protocol across all datasets and models. The evaluation metric used is recall@1, which measures the proportion of instances where the top ranked candidate exactly matches the gold-standard entity.

3. Results

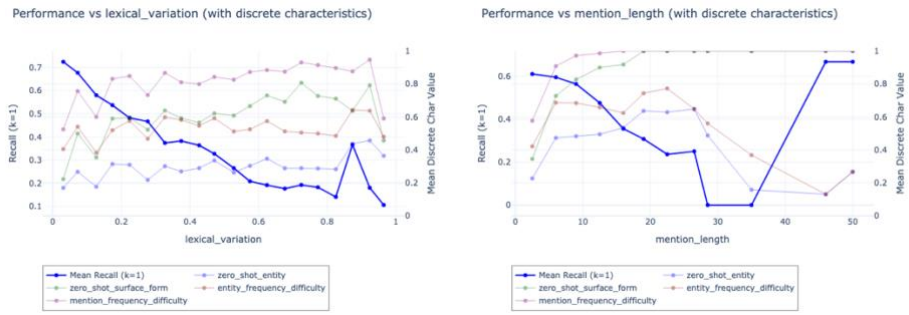
Table 1 presents the performance of ArboEL, GenBioEL, and rule-based (RBES) models across various mention characteristics introduced in Section 2.1. For each "difficult case" (e.g., high lexical variation), we report the model's performance, as well as the drop compared to the corresponding "easy" condition (e.g., low lexical variation). This helps quantify the impact of each difficulty on the models and highlights where performance degrades most significantly.

Table 1. Performance over mention characteristics ("difficult case"). The drop from the corresponding "easy" conditions are presented in red.

Characteristic	GenBioEL	ArboEL	RBES
Zero-shot entity	$0.51_{\downarrow -0.05}$	$0.48_{\downarrow -0.27}$	$0.56_{\downarrow -0.04}$
Zero-shot name	$0.50_{\downarrow -0.08}$	$0.50_{\downarrow -0.32}$	$0.48_{\downarrow -0.19}$
Long mention	$0.46_{\downarrow -0.09}$	$0.43_{\downarrow -0.26}$	$0.40_{\downarrow -0.19}$
Few synonyms	$0.54_{\downarrow -0.02}$	$0.64_{\downarrow -0.11}$	$0.53_{\downarrow -0.13}$
Homonyms exist	$0.45_{\downarrow -0.16}$	$0.68_{\downarrow -0.02}$	$0.52_{\downarrow -0.12}$
High lexical variation	$0.39_{\downarrow -0.28}$	$0.45_{\downarrow -0.42}$	$0.29_{\downarrow -0.53}$
Rare entity	$0.54_{\downarrow -0.02}$	$0.60_{\downarrow -0.17}$	$0.58_{\downarrow -0.02}$
Rare mention	$0.54_{\downarrow -0.02}$	$0.61_{\downarrow -0.23}$	$0.55_{\downarrow -0.12}$

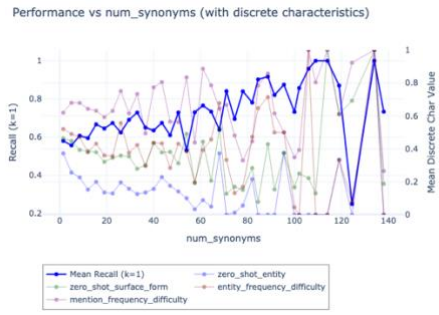
Among all characteristics, lexical variation appears to be the most challenging, with all models suffering large performance drops. This suggests that when entities are referred to using diverse and non-standard surface forms, neural and rule-based models struggle to maintain high recall. Interestingly, while GenBioEL suffers its largest

performance drop in the presence of homonyms, ArboEL is barely affected. This behavior likely comes from architectural differences: ArboEL encodes the full document context surrounding a mention, which helps disambiguate mentions, whereas GenBioEL only processes the isolated sentence containing the mention, limiting its ability to resolve such ambiguities.

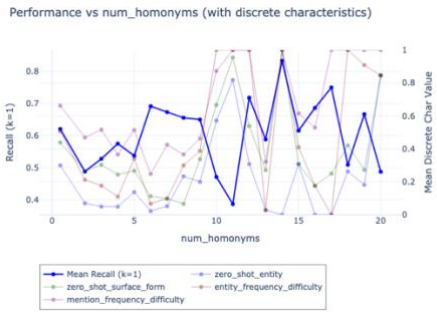


(a) Lexical variation

(b) Mention length



(c) Number of synonyms



(d) Number of homonyms

Figure 1. Performance on some characteristics, alongside the prevalence of zero-shot / rare entity and zero-shot / rare name.

4. Discussion

All models experience performance degradation under difficult conditions, but ArboEL consistently suffers the most severe drops. The most striking weaknesses of ArboEL compared to GenBioEL and RBES appear on zero-shot / rare entities or names. These consistent failures across unseen or low-frequency inputs indicate that while ArboEL performs well when entities or surface forms were seen during training, it fails to generalize. This indicates that ArboEL tends to overfit to the training data. To investigate this further, we explore whether the drop in performance on other mention features correlates with a higher proportion of unseen entities or names. Figure 1a to 1d illustrate the relationship between model performance on mention characteristics and the prevalence of zero-shot / rare entities and names. The left y-axis shows the average recall@1 of ArboEL, GenBioEL, and RBES, while the right y-axis indicates the

proportion of zero-shot / rare entities or names in MedMentions test set. It should be noted that the rightmost regions often contain very few mentions, as extreme values of the features are rare. As a result, performance becomes more unstable, noisy, and harder to interpret.

Across all these figures, we observe a decoupling effect between characteristic performance and the prevalence of unseen/rare entities or names. As model performance improves along a given characteristic, the prevalence of zero-shot / rare cases tend to decrease; conversely, when performance declines, their prevalence tends to increase. This suggests that, although certain characteristics may independently introduce difficulty, the presence of zero-shot / rare consistently underlies poor performance and is likely the primary contributing factor. These results suggest that ArboEL tends to overfit the training data, struggling particularly on unseen or rare entities and names. Although GenBioEL shows minor performance declines under difficult conditions, overall it does not outperform ArboEL, raising questions about whether its robustness reflects true generalization or simply a lower capacity to learn from training data. More broadly, this analysis highlights that some mention characteristics, such as length or lexical variation, may not be inherently difficult, but rather correlated with a lack of training coverage, suggesting that generalization remains the key challenge in biomedical entity linking.

5. Conclusion

This paper introduces a fine-grained evaluation framework for Biomedical Entity Linking that goes beyond global metrics to evaluate models. Our experiments show that although GenBioEL and ArboEL achieve strong overall results, they both struggle with specific mention characteristics, particularly lexical variation and unseen/rare entities and names. Our analysis confirms that performance degradation often correlates with lower training coverage, highlighting that generalization remains a core challenge in BEL. By extending BELB with interpretable annotations, our framework offers a valuable tool for improving biomedical entity linking and guiding future model development.

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